

Figure 18.12 Compound Epithelium

Activity 1

Rinse your mouth with water. Using a tooth pick or ice-cream stick, scrap superficial cells from inner side of the cheek and spread it on a clean glass slide. Dry the glass slide with the scrap cells taken from the inner side of cheek. Add two drops of methylene blue stain. Identify the cells under low and high power of the microscope.

18.2.2 Connective Tissues

It is one of the most abundant and widely distributed tissue. It provides **structural frame work** and gives **support** to different tissues forming organs. It prevents the organs from getting displaced by body movements.

The components of the connective tissue are the intercellular substance known as the matrix, connective tissue cells and fibres. Connective tissue is classified as follows:

- Connective tissue proper (Areolar and Adipose tissue)
- Supportive connective tissue (Cartilage and Bone)
- Dense connective tissue (Tendons and Ligaments)
- Fluid connective tissue (Blood and Lymph)

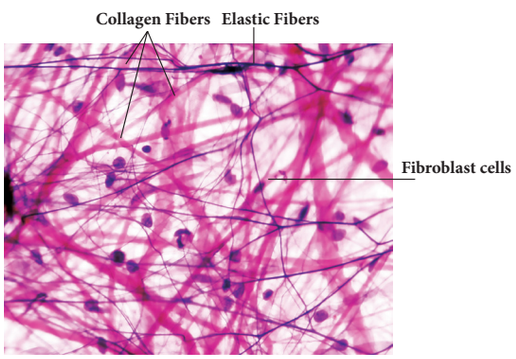


Figure 18.13 Areolar tissue

i. Connective tissue proper:

Connective tissue proper consist of collagen fibres, elastin fibres and fibroblast cells.

Areolar tissue: It has cells and fibres loosely arranged in a semi-fluid ground substance called **matrix**. It takes the form of fine threads crossing each other in every direction leaving small spaces called areolae. It joins skin to muscles, fills space inside organs and is found around muscles, blood vessels and nerves. It helps in repair of tissues after injury and fixes skin to underlying muscles.

Adipose Tissue: Adipose tissue is the aggregation of **fat cells** or **adipocytes,** spherical or oval in shape. It serves as fat reservoir.

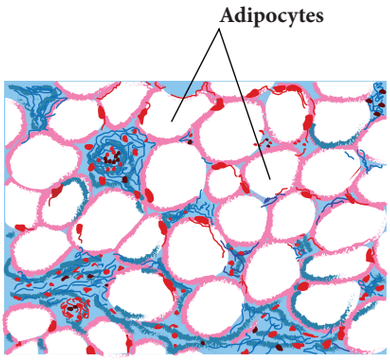


Figure 18.14 Adipose tissue

They are found in subcutaneous tissue, between internal organs around the heart and kidneys. They act as **shock absorbers** around the kidneys and eye balls. They also regulate the body temperature by acting as **insulator**.

ii. Supportive Connective Tissue:

The supporting or skeletal connective tissues forms the endoskeleton of the vertebrate body which protect various organs and help in locomotion. The supportive tissues include cartilage and bone.

Cartilage: They are soft, semi-rigid, flexible and are less vascular in nature. The matrix is composed of large cartilage cells called **chondrocytes**. These cells are present in fluid filled spaces known as **lacunae**.

IX_SCI_EM_Unit-18_BIO.indd 216 30/11/2022 20:07:47